PHYS 253: Thermal Physics

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Abstract

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1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. Grading is divided as follows:

- 10% problem sets (~ 6, submitted as handwritten PDF scans)[1]
- 50% two midterms
- 40% final exam

[1] Only one or two problems on each assignment are marked for submission and grading.

2 Temperature

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4.4.5 Heat Capacities from Equipartition

6 Entropy

As previously mentioned, Sadi Carnot was the first to recognize that heat flowing from hot to cold can be used to produce work, but never all of it. The explanation behind this limit is entropy. The second law of thermodynamics can be restated in terms of entropy:

Law 1 (The Second Law of Thermodynamics (Entropy Formulation))**.** *the entropy of an isolated system never decreases with time. As a result, isolated systems evolve toward a thermodynamic equilibrium state where the entropy is maximized.*

6.1 The Clausius Inequality

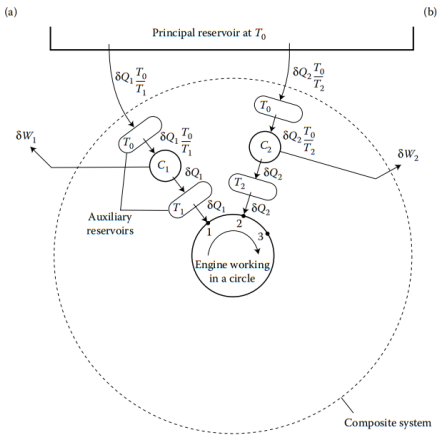


Figure 1: Clausius' thought experiment: Carnot engines route all heat exchanged by the gas through a single reservoir at T_0.

The question is: could an engine run on heat drawn from the surrounding air alone, without burning fuel? Clausius imagined a setup where a gas in a cylinder (it may be air; what matters is its temperature, not its substance) goes around an arbitrary cycle through states $1, 2, 3, \ldots$ at temperatures $T_1, T_2, T_3, \ldots$, all different from the outside air at T_0, the single principal reservoir. Going from state i to $i+1$ requires a small heat δQ_i at T_i. The idea behind his thought experiment is to consider the composite system of the gas, the Carnot engines, and the auxiliary reservoirs. In it, a Carnot engine C_i draws heat $\delta Q_i \frac{T_0}{T_i}$ from the air (via an auxiliary reservoir at T_0), converts part of it into work δW_i, and delivers the remaining δQ_i to an auxiliary reservoir at T_i, which passes it to the gas. Each auxiliary reservoir gives exactly what it receives, so it ends unchanged.

Remark 1. *The factor* $\frac{T_0}{T_i}$ *comes from the Carnot relation* $\frac{Q_{T_0}}{Q_{T_i}} = \frac{T_0}{T_i}$.[2]

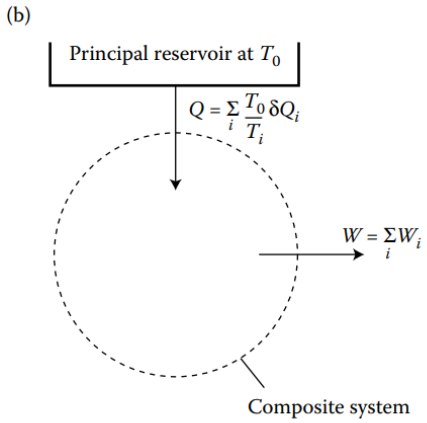


Figure 2: The composite system: gas, Carnot engines and auxiliary reservoirs, exchanging heat only with the reservoir at T_0.

We can easily better understand the result of this thought experiment by compressing Figure 1 into the composite system of Figure 2. After a full cycle everything in it is back to its original state, so $\Delta U = 0$, and the first law gives

$$W = Q = T_0 \sum_i \frac{\delta Q_i}{T_i}, \tag{1}$$

where $W = \sum_i \delta W_i$ is the total work done and Q the heat drawn from T_0. A positive $W = Q$ would mean heat from a single reservoir fully converted into work, forbidden by:

Law 2 (Kelvin Statement of the Second Law of Thermodynamics)**.** *No process is possible whose sole result is the absorption of heat from a reservoir and the conversion of this heat into work.*

Hence $W = Q \leq 0$, and since $T_0 > 0$, $\sum_i \frac{\delta Q_i}{T_i} \leq 0$. Taking the steps infinitesimal turns the sum into an integral:

Law 3 (Clausius Inequality)**.** *For any cyclic process,*

$$\oint \frac{\delta Q}{T} \leq 0, \tag{2}$$

where T *is the temperature at which* δQ *is supplied, with equality if and only if the cycle is reversible.*

[2] For example, with $T_0 = 600$ K, $T_1 = 300$ K and $\delta Q_1 = 10$ J, the Carnot engine takes 20 J from the air, delivers 10 J to the gas, and the difference, 10 J, is the work δW_1.

6.2 Entropy

6.2.1 Entropy as a State Function

6.2.2 Entropy Changes in Irreversible Processes

6.3 The Principle of Increasing Entropy

6.3.1 Isolated Systems

6.3.2 Entropy of the Universe

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6.4 Entropy–Temperature Diagrams

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6.6 Other Examples of Entropy Calculations

6.6.1 Free Expansion

6.6.2 Measuring Entropies

7 Statistical Mechanics

7.1 Introduction to Probability and Statistics

Definition 1 (Microstate)**.** *A microstate is a specific configuration of the particles in a system.*

Definition 2 (Macrostate)**.** *A macrostate is a description of the system in terms of macroscopic properties, such as temperature, pressure, and volume. Different macrostates can be distinguished by a measurement, while different microstates of the same macrostate cannot.*

Assuming all microstates are equally probable, the probability of any macrostate is given by the multiplicity of that macrostate divided by the sum of all the multiplicities:

$$P(n) = \frac{\Omega(n)}{\sum_i \Omega(i)}. \tag{3}$$

For N two-state particles, the multiplicity $\Omega(N, n)$ of the macrostate with n out of the N particles in one of the two states can be calculated using the binomial coefficient:

$$\Omega(N, n) = \binom{N}{n} = \frac{N!}{n!(N-n)!}. \tag{4}$$

There are $N + 1$ macrostates and 2^N microstates in total:

$$\sum_{n=0}^{N} \binom{N}{n} = 2^N. \tag{5}$$

The distribution has mean $N/2$ and standard deviation $\sigma = \sqrt{N}/2$, so its relative width is

$$\frac{\sigma}{N} = \frac{1}{2\sqrt{N}}, \tag{6}$$

meaning the multiplicity becomes sharply peaked at $n = N/2$ as N grows.

For $N \gg 1$, factorials are evaluated with Stirling's approximation:

$$N! \approx N^N e^{-N} \sqrt{2\pi N}, \qquad \ln N! \approx N \ln N - N. \tag{7}$$

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